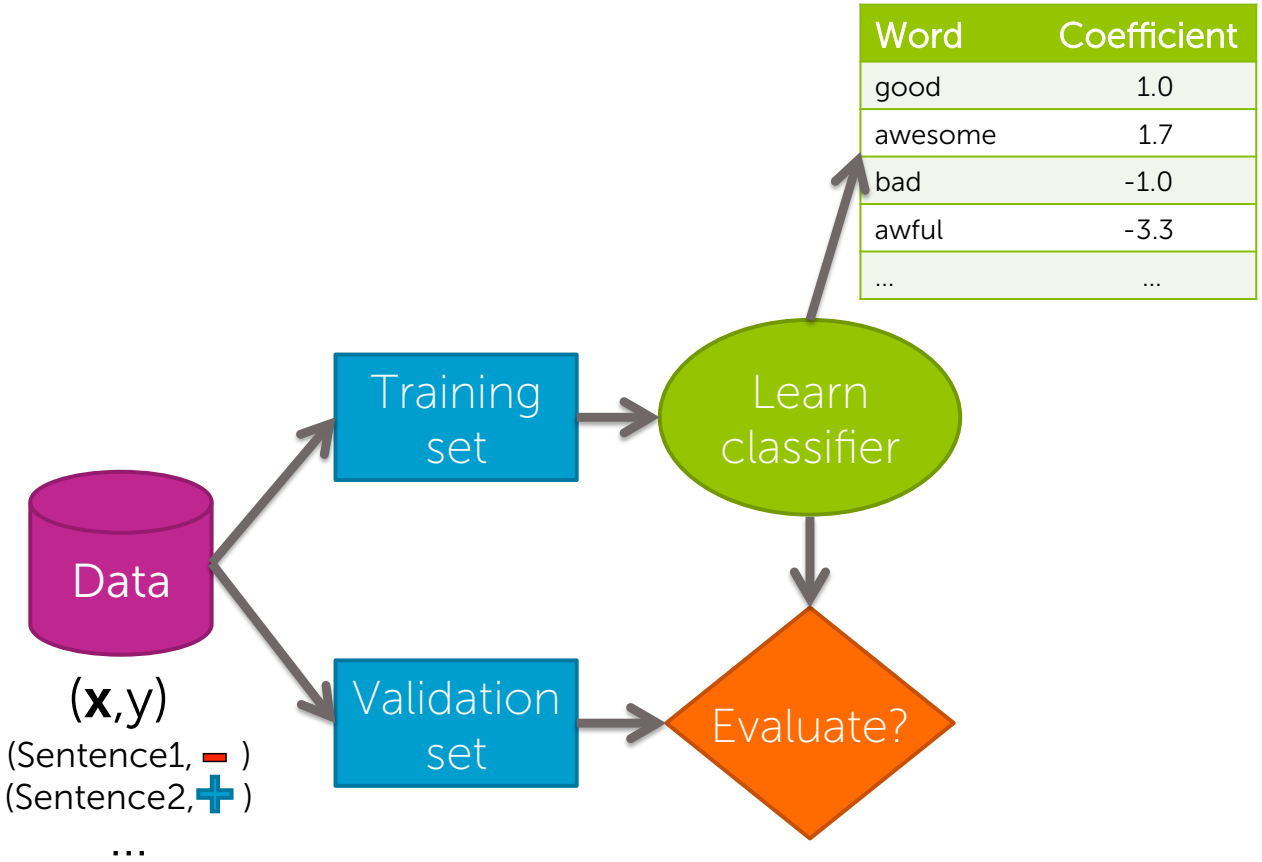
Linear classifiers:
Overfitting & regularization

©2015-2016 Emily Fox & Carlos Guestrin

Training and evaluating a classifier

©2015-2016 Emily Fox & Carlos Guestrin

Training a classifier = Learning the coefficients



Word	Coefficient
good	1.0
awesome	1.7
bad	-1.0
awful	-3.3
...	...

©2015-2016 Emily Fox & Carlos Guestrin

Classification error

Learned classifier

$\hat{y}$ =

Test example

(Sushi was great,)

Mistake!

Correct	0
Mistakes	1

Hide label

©2015-2016 Emily Fox & Carlos Guestrin

Classification error & accuracy

- Error measures fraction of mistakes

$$\text{error} = \frac{\#\ \text{Mistakes}}{\text{Total number of data points}}$$

  – Best possible value is 0.0

- Often, measure **accuracy**
  – Fraction of correct predictions

$$\text{accuracy} = \frac{\#\ \text{Correct}}{\text{Total number of data points}}$$

  – Best possible value is 1.0

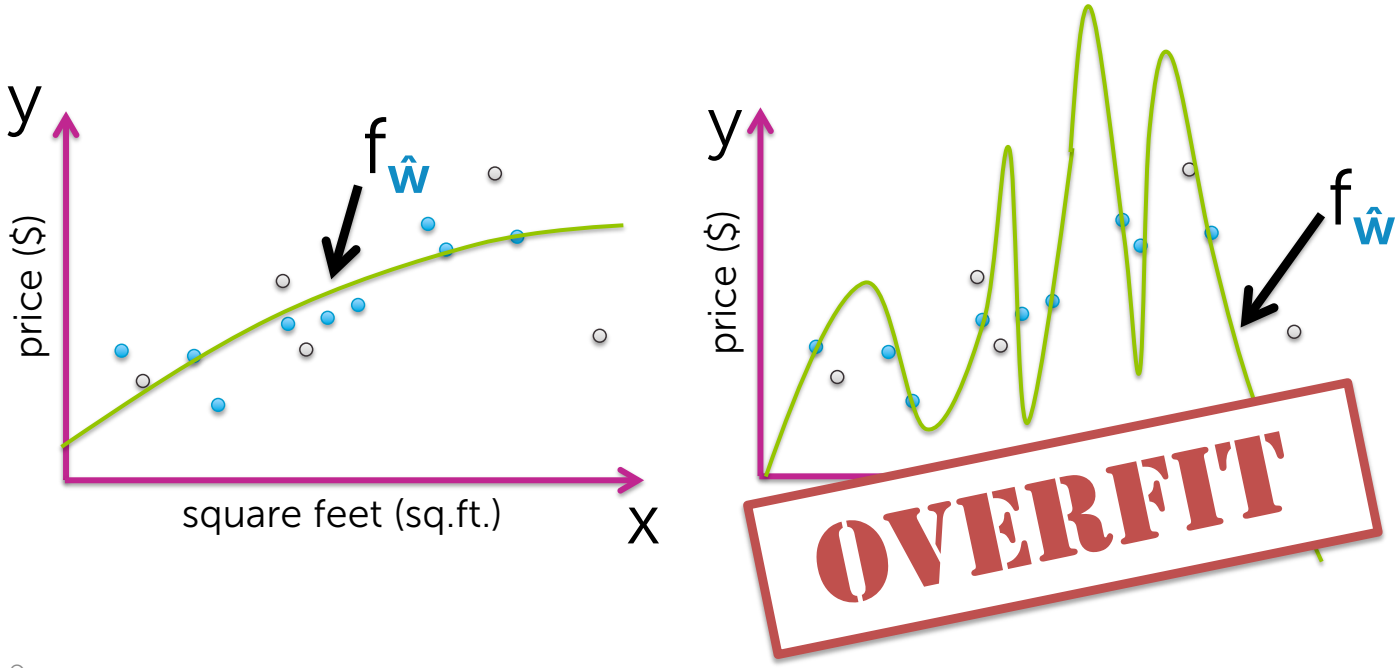
©2015-2016 Emily Fox & Carlos Guestrin

Overfitting in regression: review

©2015-2016 Emily Fox & Carlos Guestrin

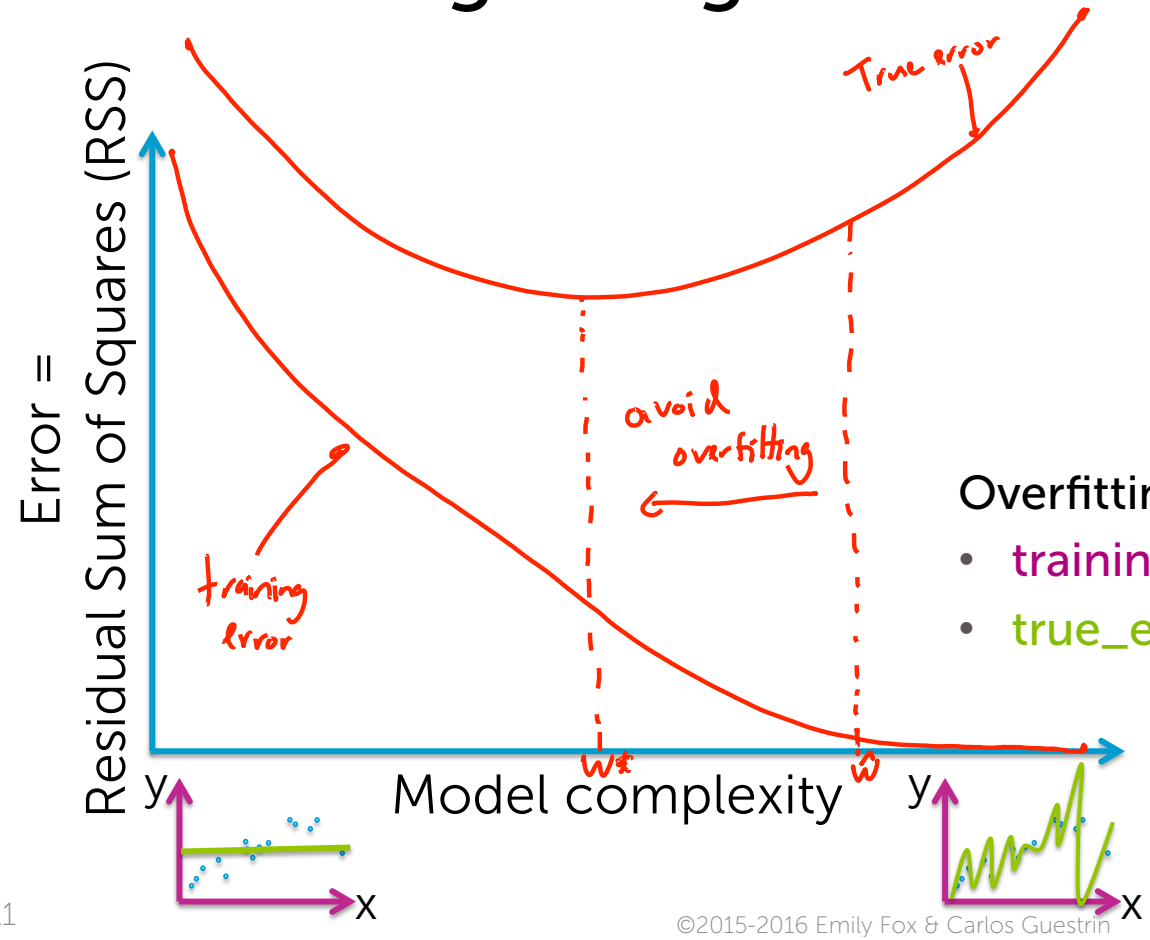
Flexibility of high-order polynomials

$$y_i = w_0 + w_1 x_i + w_2 x_i^2 + \ldots + w_p x_i^p + \varepsilon_i$$



©2015-2016 Emily Fox & Carlos Guestrin

Overfitting in regression



Overfitting if there exists **w***:

- training_error(**w***) > training_error(**ŵ**)
- true_error(**w***) < true_error(**ŵ**)

©2015-2016 Emily Fox & Carlos Guestrin

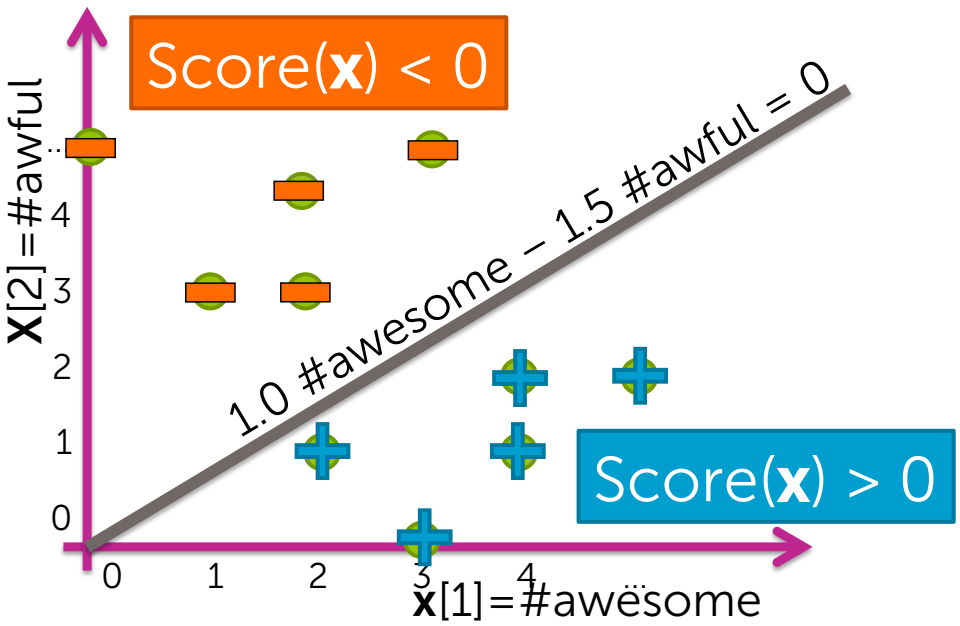
Overfitting in classification

©2015-2016 Emily Fox & Carlos Guestrin

Decision boundary example

Word	Coefficient
#awesome	1.0
#awful	-1.5

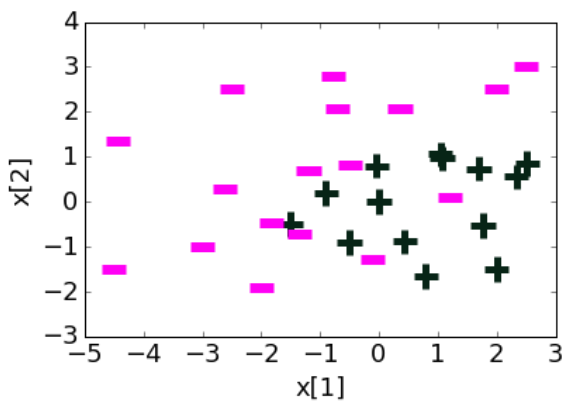
Score(x) = 1.0 #awesome – 1.5 #awful

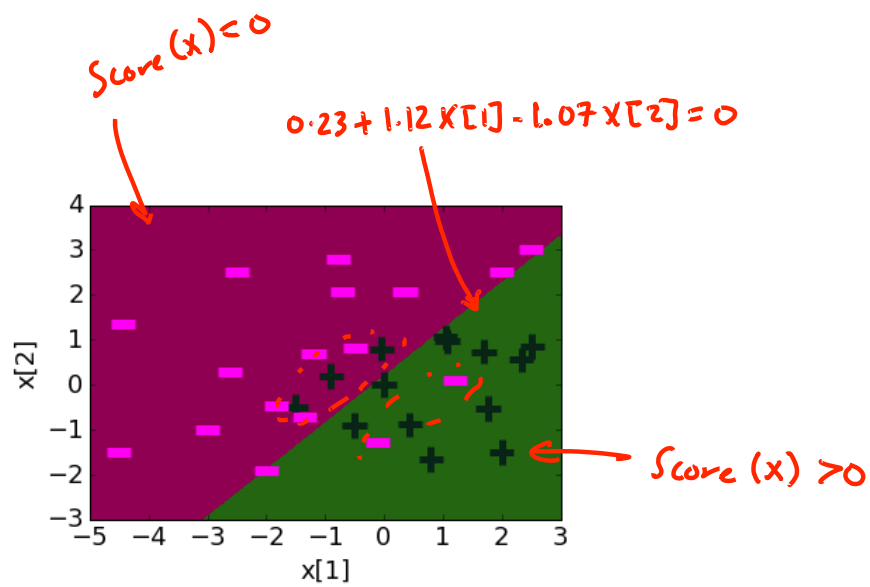


©2015-2016 Emily Fox & Carlos Guestrin

Learned decision boundary

Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	w_0 1	0.23
$h_1(\mathbf{x})$	w_1 $\mathbf{x}[1]$	1.12
$h_2(\mathbf{x})$	w_2 $\mathbf{x}[2]$	−1.07



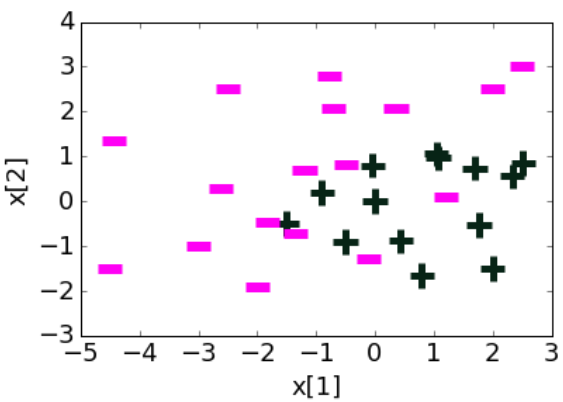


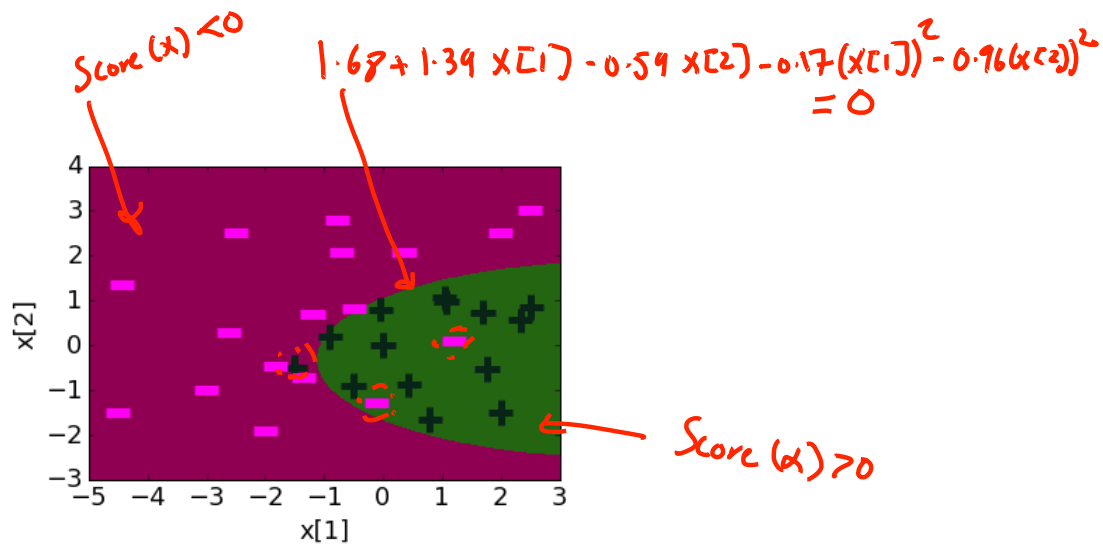
©2015-2016 Emily Fox & Carlos Guestrin

Quadratic features (in 2d)

Note: we are not including cross terms for simplicity

Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	1	1.68
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	1.39
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	-0.59
$h_3(\mathbf{x})$	$(\mathbf{x}[1])^2$	-0.17
$h_4(\mathbf{x})$	$(\mathbf{x}[2])^2$	-0.96





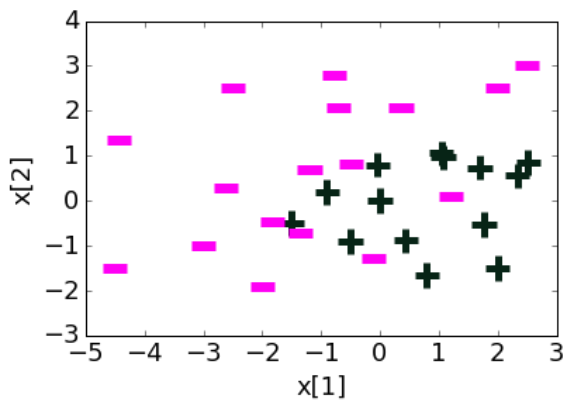
©2015-2016 Emily Fox & Carlos Guestrin

Degree 6 features (in 2d)

Note: we are not including cross terms for simplicity

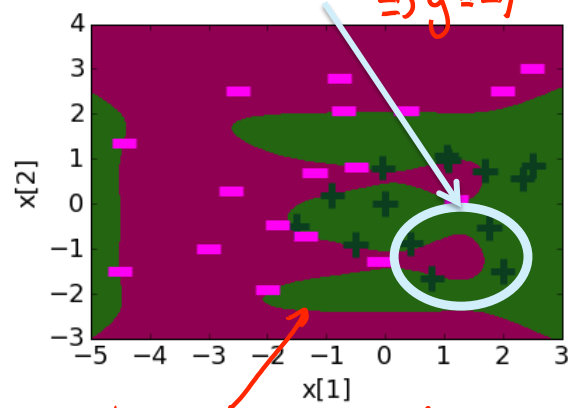
Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	1	21.6
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	5.3
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	-42.7
$h_3(\mathbf{x})$	$(\mathbf{x}[1])^2$	-15.9
$h_4(\mathbf{x})$	$(\mathbf{x}[2])^2$	-48.6
$h_5(\mathbf{x})$	$(\mathbf{x}[1])^3$	-11.0
$h_6(\mathbf{x})$	$(\mathbf{x}[2])^3$	67.0
$h_7(\mathbf{x})$	$(\mathbf{x}[1])^4$	1.5
$h_8(\mathbf{x})$	$(\mathbf{x}[2])^4$	48.0
$h_9(\mathbf{x})$	$(\mathbf{x}[1])^5$	4.4
$h_{10}(\mathbf{x})$	$(\mathbf{x}[2])^5$	-14.2
$h_{11}(\mathbf{x})$	$(\mathbf{x}[1])^6$	0.8
$h_{12}(\mathbf{x})$	$(\mathbf{x}[2])^6$	-8.6

Coefficient values getting large



Score($\mathbf{x}$) < 0

$\Rightarrow \hat{y} = -1$



extremely complex (crazy) decision boundary

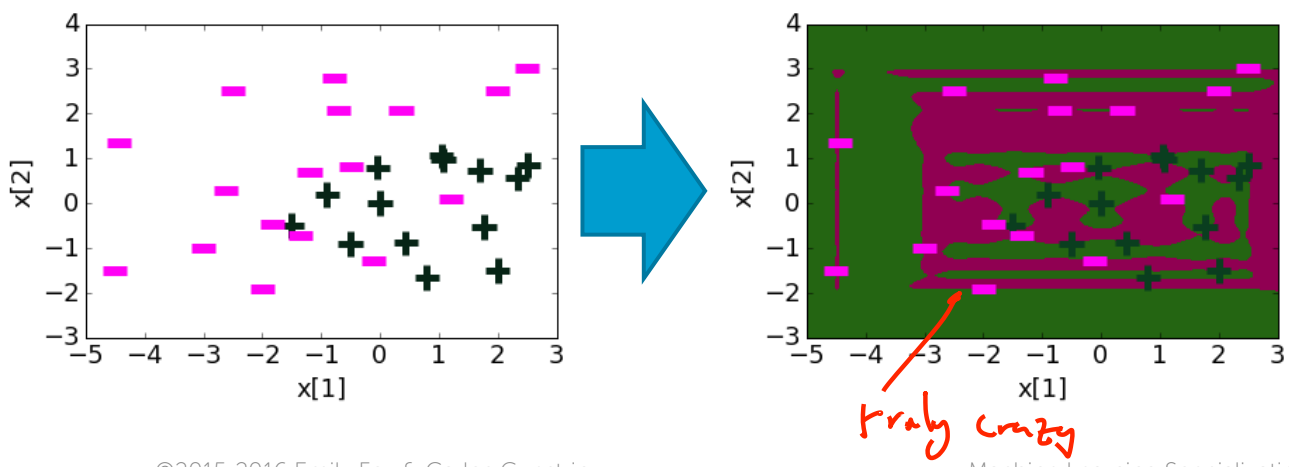
©2015-2016 Emily Fox & Carlos Guestrin

Degree 20 features (in 2d)

Note: we are not including cross terms for simplicity

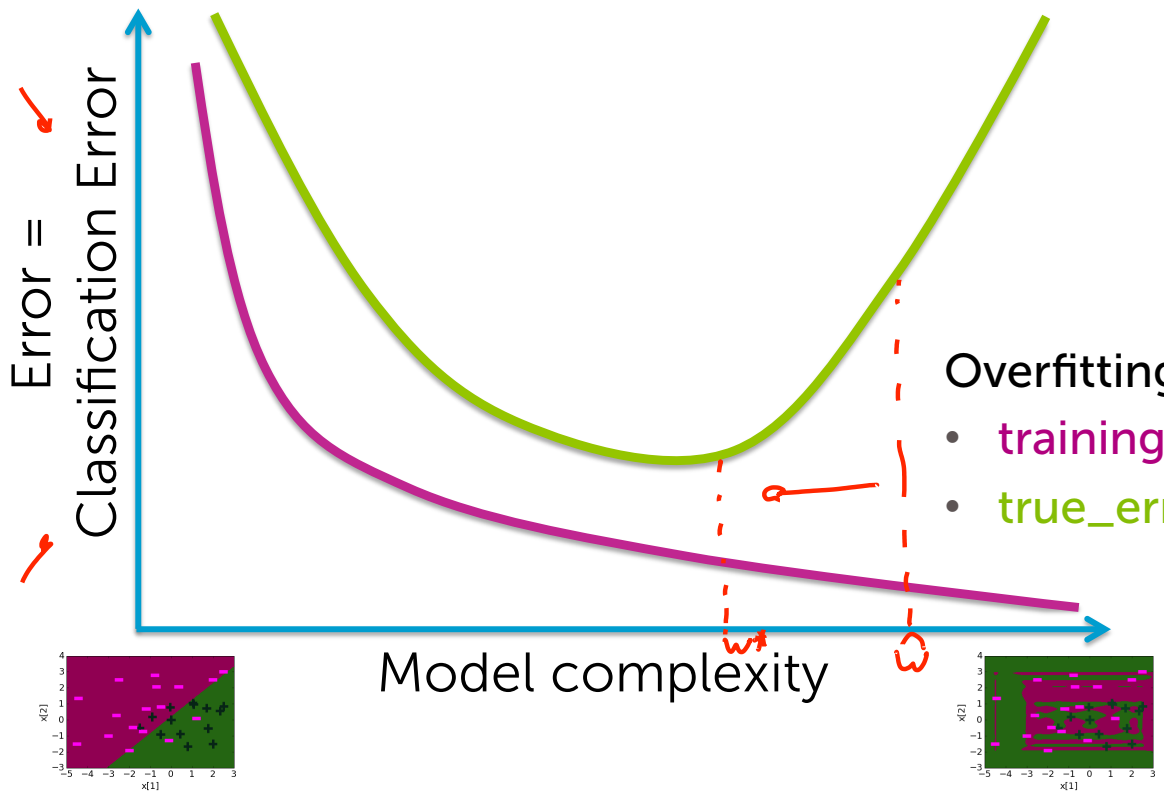
Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	1	8.7
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	5.1
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	78.7
…	…	…
$h_{11}(\mathbf{x})$	$(\mathbf{x}[1])^6$	-7.5
$h_{12}(\mathbf{x})$	$(\mathbf{x}[2])^6$	**3803**
$h_{13}(\mathbf{x})$	$(\mathbf{x}[1])^7$	~~21.1~~
$h_{14}(\mathbf{x})$	$(\mathbf{x}[2])^7$	**-2406**
…	…	~~…~~
$h_{37}(\mathbf{x})$	$(\mathbf{x}[1])^{19}$	$-2*10^{-6}$
$h_{38}(\mathbf{x})$	$(\mathbf{x}[2])^{19}$	-0.15
$h_{39}(\mathbf{x})$	$(\mathbf{x}[1])^{20}$	$-2*10^{-8}$
$h_{40}(\mathbf{x})$	$(\mathbf{x}[2])^{20}$	0.03

Often, overfitting associated with very large estimated coefficients $\hat{\mathbf{w}}$



©2015-2016 Emily Fox & Carlos Guestrin

Overfitting in classification



Overfitting if there exists **w***:

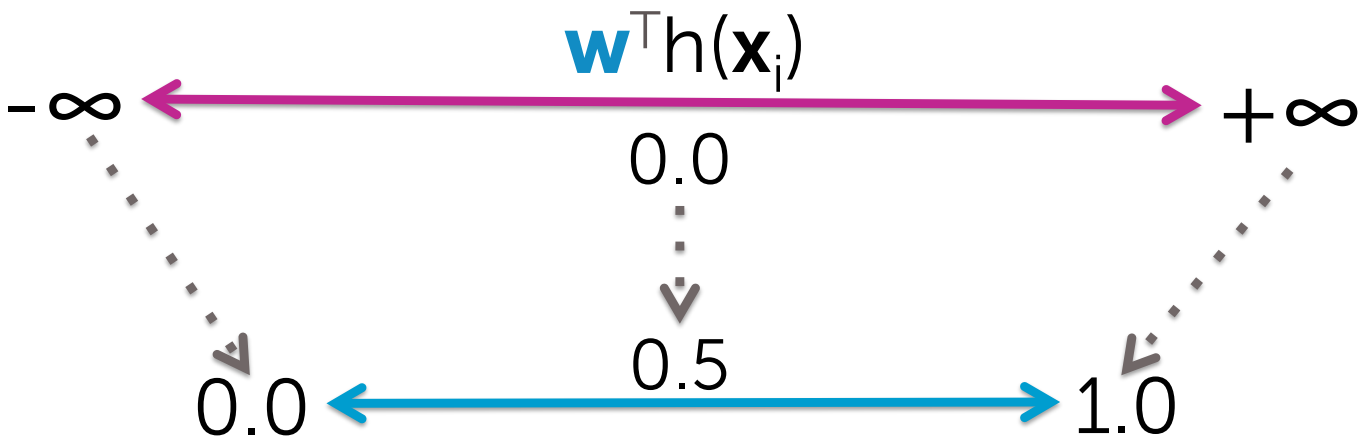
- training_error(**w***) > training_error(**ŵ**)
- true_error(**w***) < true_error(**ŵ**)

©2015-2016 Emily Fox & Carlos Guestrin

Overfitting in classifiers → Overconfident predictions

©2015-2016 Emily Fox & Carlos Guestrin

Logistic regression model



$$P(y=+1|\mathbf{x}_i,\mathbf{w}) = \text{sigmoid}(\mathbf{w}^T h(\mathbf{x}_i))$$

©2015-2016 Emily Fox & Carlos Guestrin

The subtle (negative) consequence of overfitting in logistic regression

Overfitting ➔ Large coefficient values

$\hat{\mathbf{w}}^T h(\mathbf{x}_i)$ is very positive (or very negative)
➔ **sigmoid**($\hat{\mathbf{w}}^T h(\mathbf{x}_i)$) goes to 1 (or to 0)

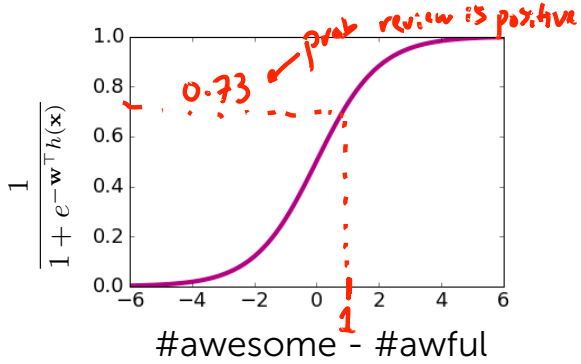
Model becomes extremely overconfident of predictions

©2015-2016 Emily Fox & Carlos Guestrin

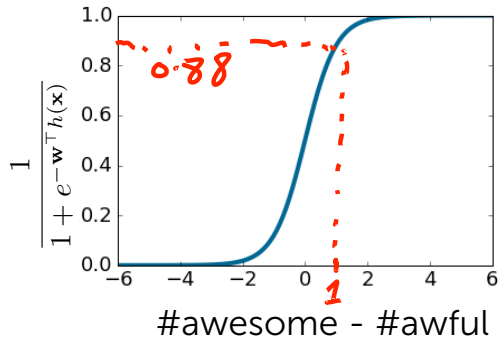
Effect of coefficients on logistic regression model

Input x: #awesome=2, #awful=1

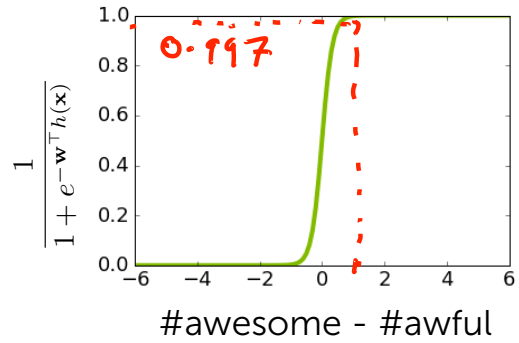
w_0	0
$w_{\#awesome}$	+1
$w_{\#awful}$	-1



w_0	0
$w_{\#awesome}$	+2
$w_{\#awful}$	-2



w_0	0
$w_{\#awesome}$	+6
$w_{\#awful}$	-6



©2015-2016 Emily Fox & Carlos Guestrin

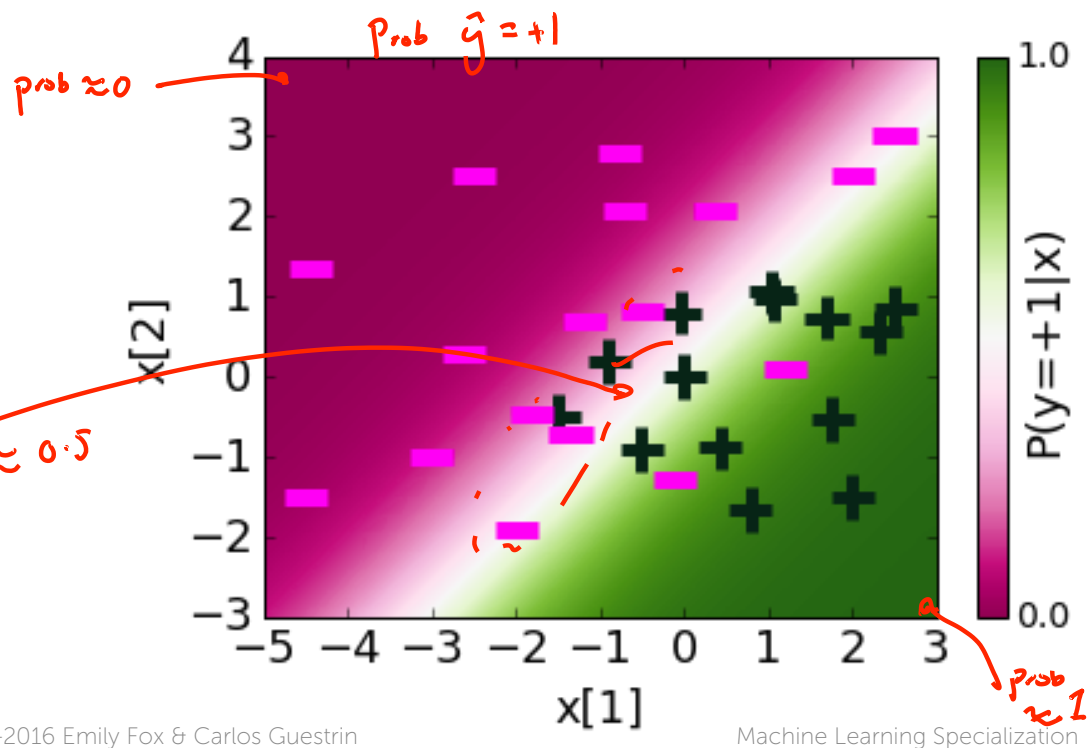
Learned probabilities

Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	1	0.23
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	1.12
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	-1.07

$$P(y = +1 \mid \mathbf{x}, \mathbf{w}) = \frac{1}{1 + e^{-\mathbf{w}^\top h(\mathbf{x})}}$$

Make sense

wide region of uncertainty



©2015-2016 Emily Fox & Carlos Guestrin

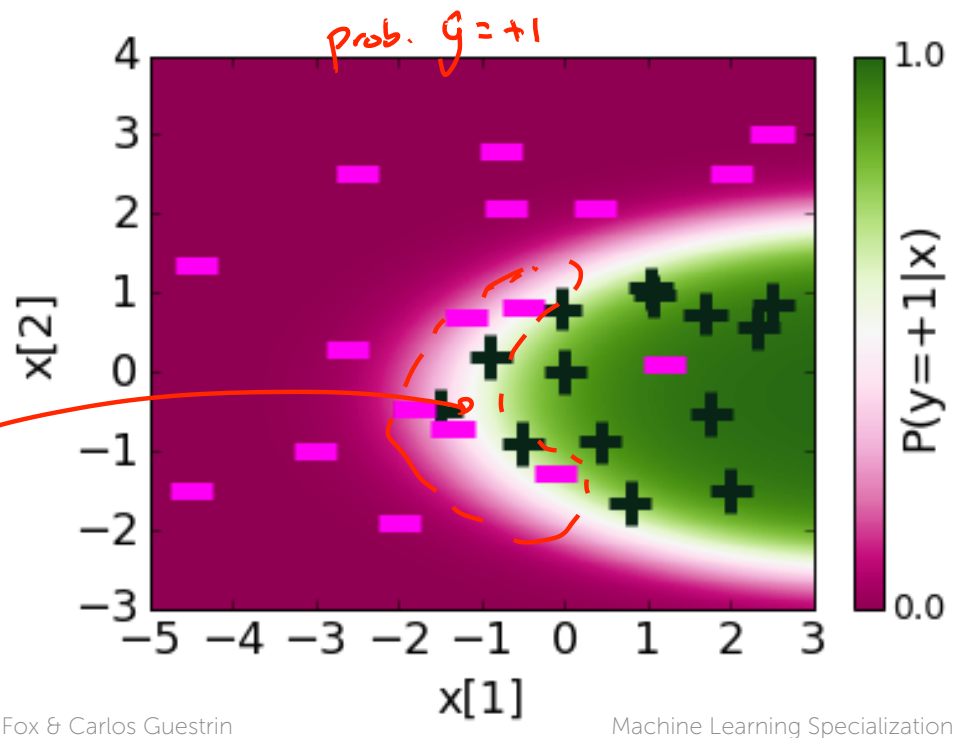
Quadratic features: Learned probabilities

Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	1	1.68
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	1.39
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	-0.58
$h_3(\mathbf{x})$	$(\mathbf{x}[1])^2$	-0.17
$h_4(\mathbf{x})$	$(\mathbf{x}[2])^2$	-0.96

better fit to data

$$P(y=+1 \mid \mathbf{x}, \mathbf{w}) = \frac{1}{1+e^{-\mathbf{w}^\top h(\mathbf{x})}}$$

uncertainty region narrower

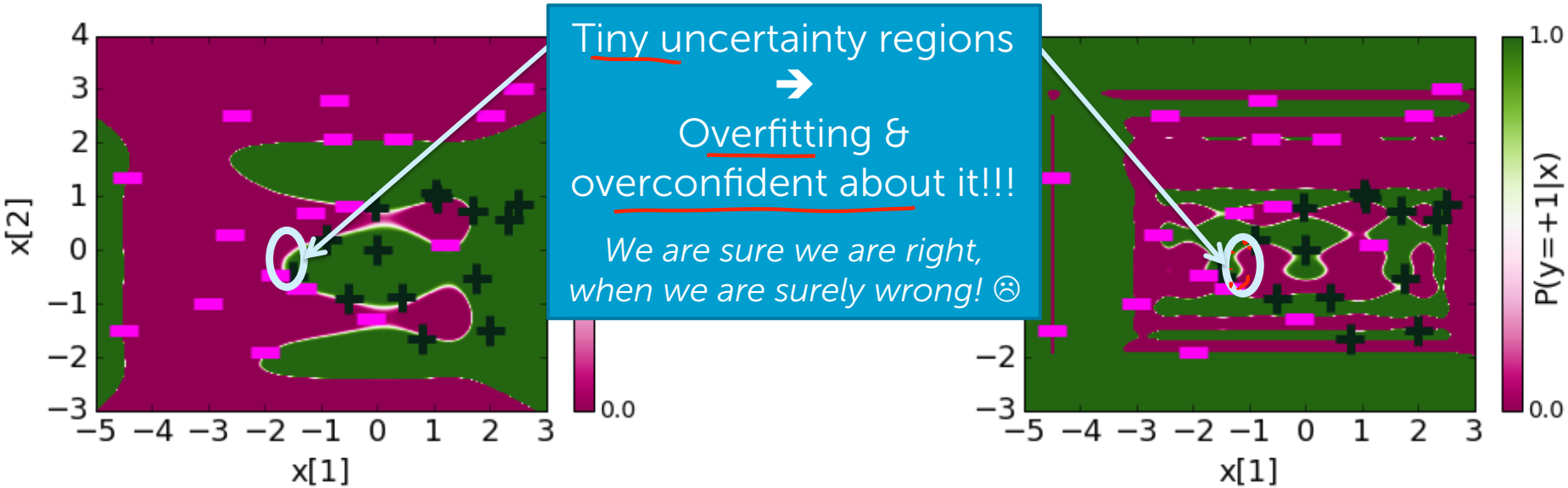


©2015-2016 Emily Fox & Carlos Guestrin

Overfitting ➔ Overconfident predictions

Degree 6: Learned probabilities

Degree 20: Learned probabilities



Tiny uncertainty regions
➔
Overfitting & overconfident about it!!!

We are sure we are right, when we are surely wrong! ☹

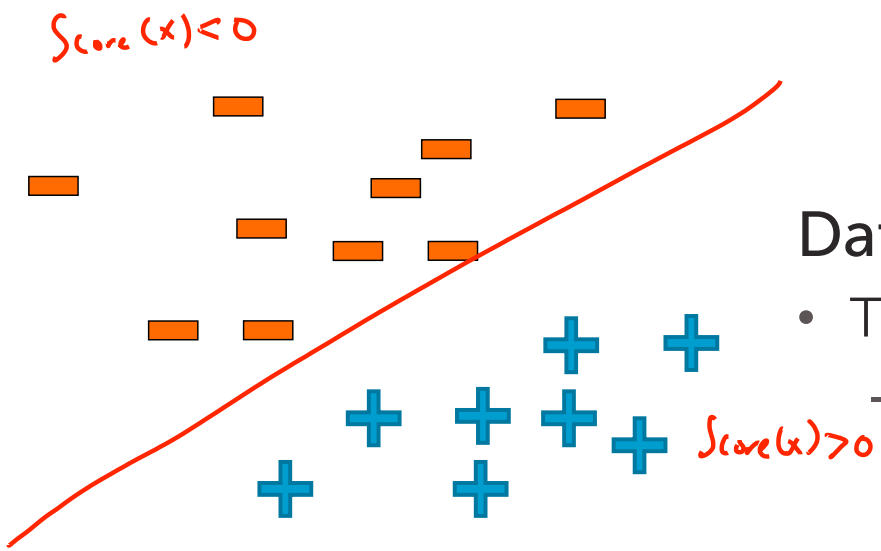
©2015-2016 Emily Fox & Carlos Guestrin

Overfitting in logistic regression: Another perspective

OPTIONAL

©2015-2016 Emily Fox & Carlos Guestrin

Linearly-separable data



Data are linearly separable if:

- There exist coefficients $\hat{\mathbf{w}}$ such that:
  - For all positive training data

    $\text{Score}(x) = \hat{w}^T h(x) > 0$
  - For all negative training data

    $\text{Score}(x) = \hat{w}^T h(x) < 0$

Note 1: If you are using D features, linear separability happens in a D-dimensional space

Note 2: If you have enough features, data are (almost) always linearly separable

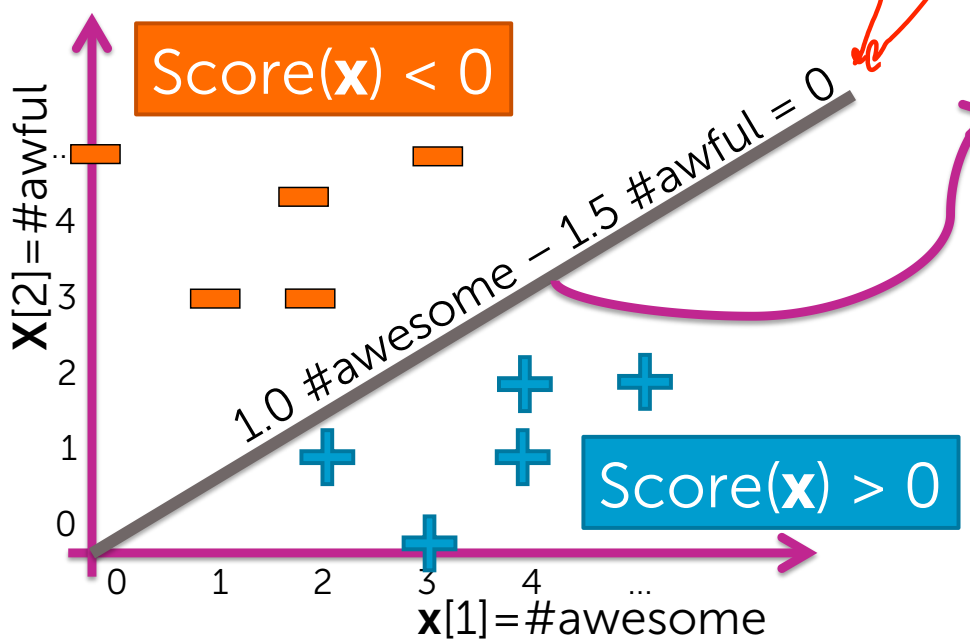
$$\text{training_error}(\hat{\mathbf{w}}) = 0$$

©2015-2016 Emily Fox & Carlos Guestrin

Effect of linear separability on coefficients

1.0 #awesome – 1.5 #awful = 0

10 #awesome – 15 #awful = 0



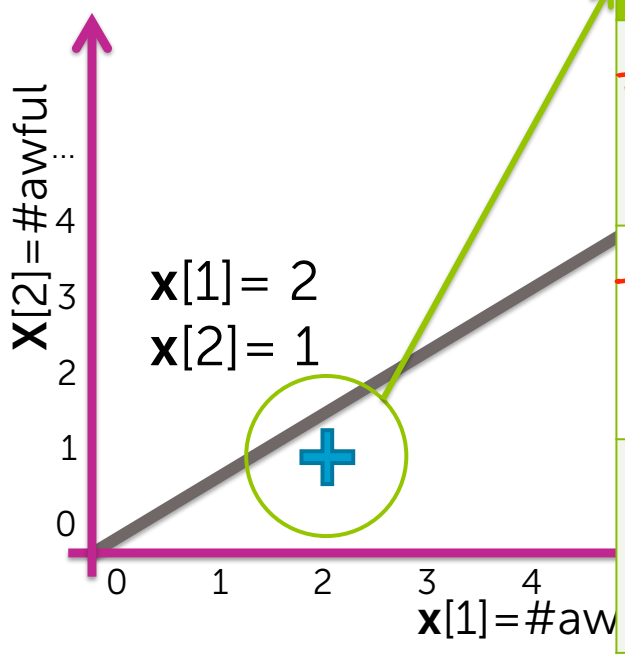
Data are linearly separable with $\hat{w}_1$=1.0 and $\hat{w}_2$=-1.5

Data also linearly separable with $\hat{w}_1$=10 and $\hat{w}_2$=-15

Data also linearly separable with $\hat{w}_1$=10^9 and $\hat{w}_2$=-1.5x10^9

©2015-2016 Emily Fox & Carlos Guestrin

Maximum likelihood estimation (MLE)
prefers most certain model ➔
Coefficients go to infinity for linearly-separable data!!!



$P(y=+1\|\mathbf{x}_i, \hat{\mathbf{w}})$
$\frac{1}{1+e^{-(2\cdot 1.5)}} = \frac{1}{1+e^{-0.5}} = 0.62$
$\frac{1}{1+e^{-5}} = 0.99$
$\frac{1}{1+e^{-0.3\,10^9}} \approx 1$

Data is linearly separable with $\hat{w}_1$=1.0 and $\hat{w}_2$=-1.5

Data also linearly separable with $\hat{w}_1$=10 and $\hat{w}_2$=-15

Data also linearly separable with $\hat{w}_1$=10^9 and $\hat{w}_2$=-1.5x10^9

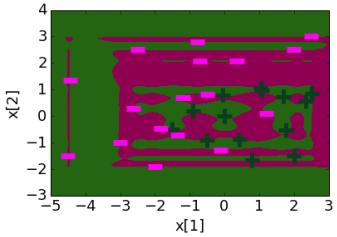
©2015-2016 Emily Fox & Carlos Guestrin

Overfitting in logistic regression is "twice as bad"

Learning tries to find decision boundary that separates data

If data are linearly separable

Overly complex boundary



Coefficients go to infinity!

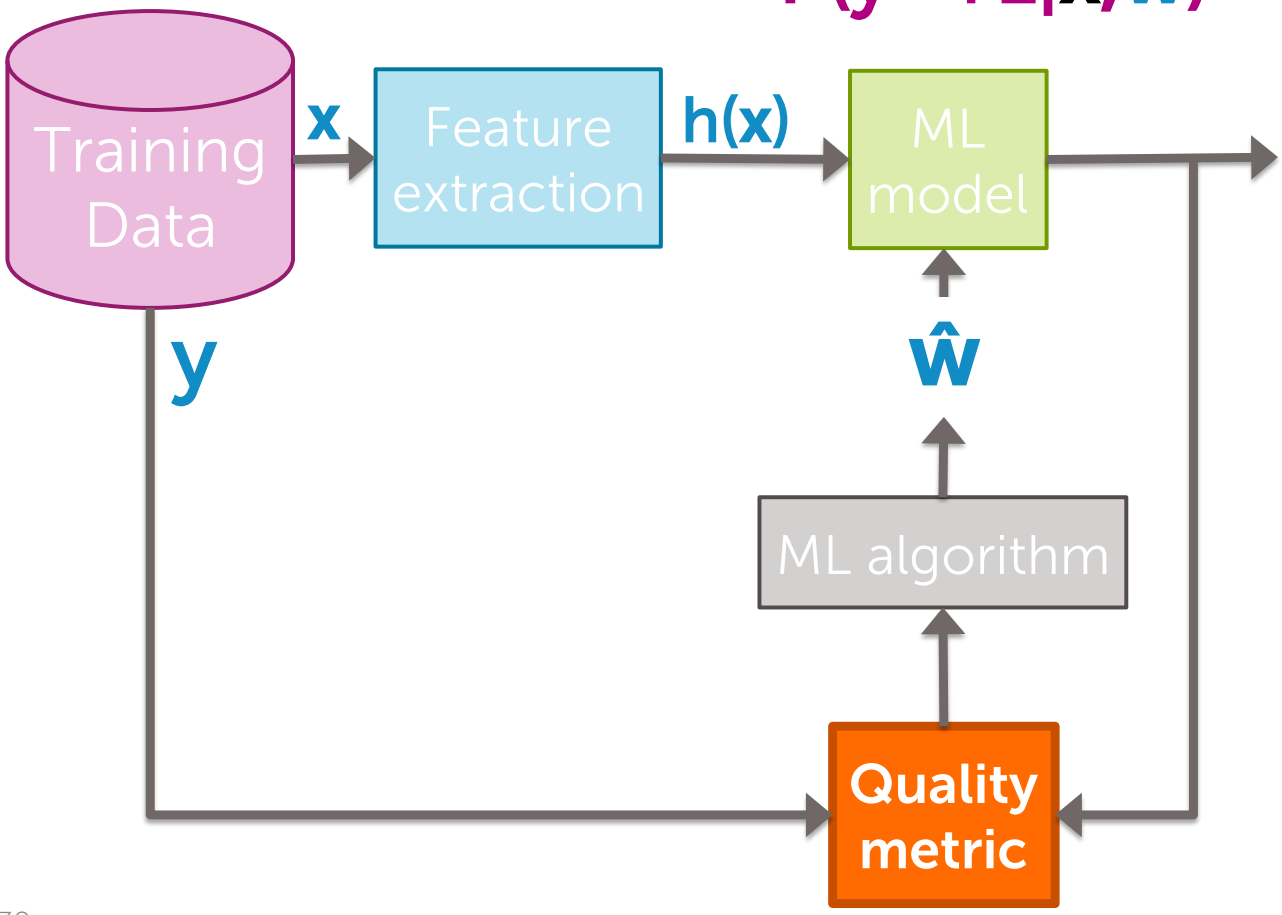
$\hat{\mathbf{w}}_1 = 10^9$
$\hat{\mathbf{w}}_2 = -1.5 \times 10^9$

©2015-2016 Emily Fox & Carlos Guestrin

Penalizing large coefficients to mitigate overfitting

©2015 Emily Fox & Carlos Guestrin

$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1+e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$



©2015-2016 Emily Fox & Carlos Guestrin

Desired total cost format

Want to balance:

i. How well function fits data

ii. Magnitude of coefficients

want to balance

Total quality =

measure of fit - measure of magnitude of coefficients

(data likelihood)
large # = good fit to training data

large # = overfit

©2015 Emily Fox & Carlos Guestrin

Maximum likelihood estimation (MLE): Measure of fit = Data likelihood

- Choose coefficients **w** that maximize likelihood:

$$\prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

- Typically, we use the log of likelihood function
  (simplifies math and has better convergence properties)

$$\ell(\mathbf{w}) = \ln \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

©2015-2016 Emily Fox & Carlos Guestrin

Measure of magnitude of logistic regression coefficients

What summary # is indicative of size of logistic regression coefficients?

- Sum of squares (L_2 norm)

$$\|w\|_2^2 = w_0^2 + w_1^2 + w_2^2 + \cdots + w_D^2$$

Penalize large coefficients

- Sum of absolute value (L_1 norm)

$$\|w\|_1 = |w_0| + |w_1| + |w_2| + \cdots + |w_D|$$

Sparse solution

©2015 Emily Fox & Carlos Guestrin

Consider specific total cost

$\max_{\mathbf{w}}$

Total quality =

measure of fit - measure of magnitude of coefficients

$$\underbrace{\ell(\mathbf{w})}_{\text{log data likelihood}} \qquad \underbrace{||\mathbf{w}||_2^2}_{L_2 \text{ penalty}}$$

©2015 Emily Fox & Carlos Guestrin

Consider resulting objective

What if $\hat{\mathbf{w}}$ selected to minimize

$$\ell(\mathbf{w}) - \lambda||\mathbf{w}||_2^2$$

tuning parameter = balance of fit and magnitude

If λ=0:

Reduces $\max_w \ell(w)$ → Standard (unpenalized) MLE solution

If λ=∞:

$\max_w \ell(w) - \infty ||w||_2^2$ → only care about penalizing w, large coefficients → $w=0$

If λ in between:

Balance data fit against the magnitude of the coefficients

©2015 Emily Fox & Carlos Guestrin

Consider resulting objective

What if $\hat{\mathbf{w}}$ selected to minimize

$$\ell(\mathbf{w}) - \lambda||\mathbf{w}||_2^2$$

tuning parameter = balance of fit and magnitude

L_2 regularized logistic regression

Pick λ using:
- Validation set (for large datasets)
- Cross-validation (for smaller datasets)

(see regression course)

©2015 Emily Fox & Carlos Guestrin

Bias-variance tradeoff

Large λ:

high bias, low variance

(e.g., $\hat{\mathbf{w}}$ =0 for $\lambda=\infty$)

In essence, λ controls model complexity

Small λ:

low bias, high variance

(e.g., maximum likelihood (MLE) fit of high-order polynomial for λ=0)

©2015 Emily Fox & Carlos Guestrin

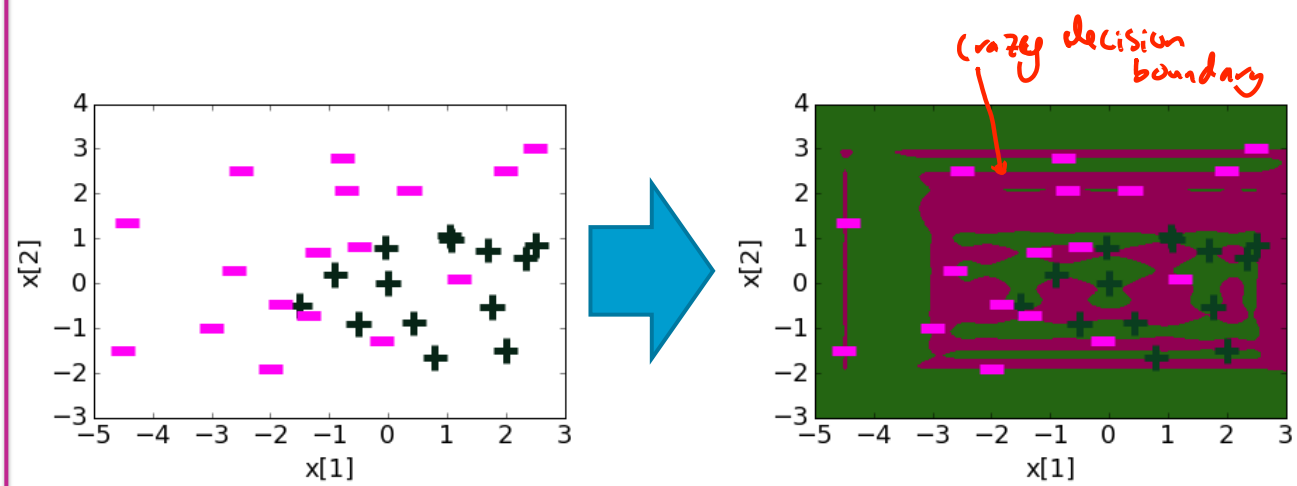
Visualizing effect of regularization on logistic regression

©2015-2016 Emily Fox & Carlos Guestrin

Degree 20 features, λ=0

Feature	Value	Coefficient learned
$h_0(\mathbf{x})$	1	8.7
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	5.1
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	78.7
…	…	…
$h_{11}(\mathbf{x})$	$(\mathbf{x}[1])^6$	-7.5
$h_{12}(\mathbf{x})$	$(\mathbf{x}[2])^6$	3803
$h_{13}(\mathbf{x})$	$(\mathbf{x}[1])^7$	21.1
$h_{14}(\mathbf{x})$	$(\mathbf{x}[2])^7$	-2406
…	…	…
$h_{37}(\mathbf{x})$	$(\mathbf{x}[1])^{19}$	$-2*10^{-6}$
$h_{38}(\mathbf{x})$	$(\mathbf{x}[2])^{19}$	-0.15
$h_{39}(\mathbf{x})$	$(\mathbf{x}[1])^{20}$	$-2*10^{-8}$
$h_{40}(\mathbf{x})$	$(\mathbf{x}[2])^{20}$	0.03

Coefficients range from **-3170** to **3803**

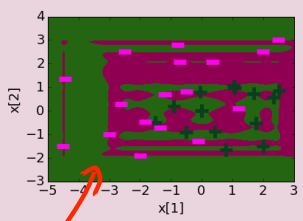
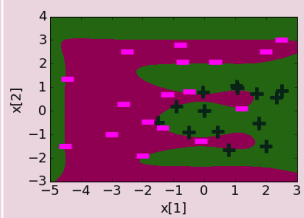
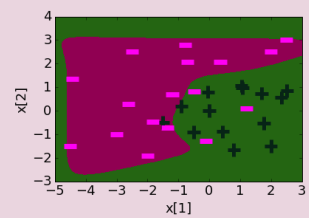
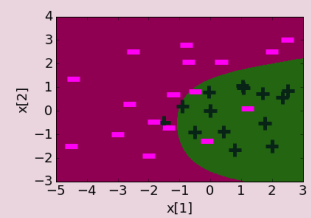
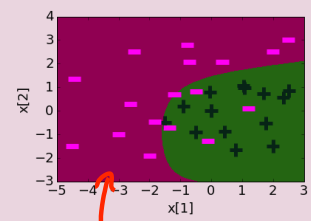


©2015-2016 Emily Fox & Carlos Guestrin

Degree 20 features, effect of regularization penalty λ

very large

smaller coefficients

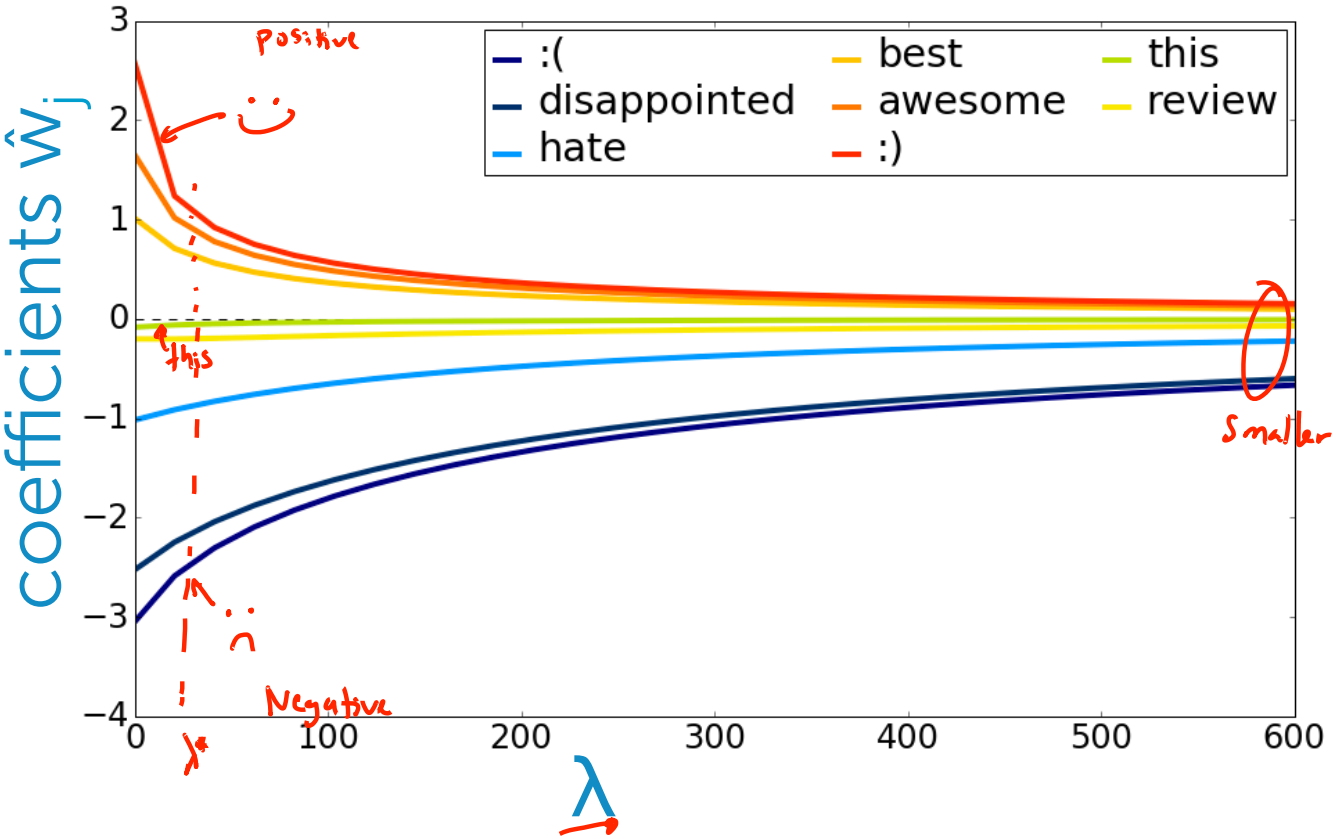
Regularization	λ = 0	λ = 0.00001	λ = 0.001	λ = 1	λ = 10
Range of coefficients	-3170 to 3803	-8.04 to 12.14	-0.70 to 1.25	-0.13 to 0.57	-0.05 to 0.22
Decision boundary					

crazy decision boundary

nicer smoother

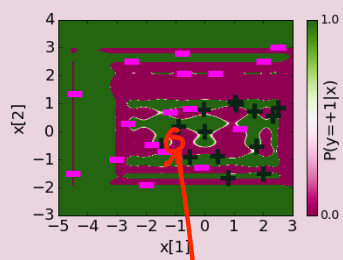
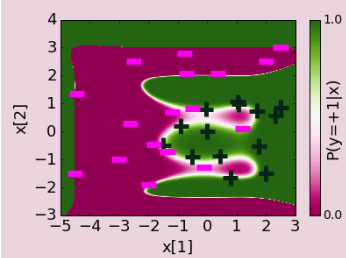
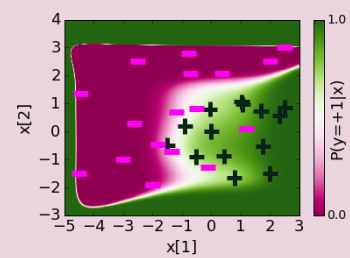
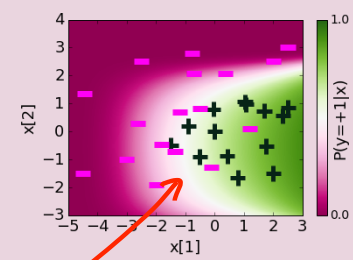
©2015-2016 Emily Fox & Carlos Guestrin

Coefficient path



©2015 Emily Fox & Carlos Guestrin

Degree 20 features: regularization reduces "overconfidence"

Regularization	λ = 0	λ = 0.00001	λ = 0.001	λ = 1
Range of coefficients	-3170 to 3803	-8.04 to 12.14	-0.70 to 1.25	-0.13 to 0.57
Learned probabilities				

highly over confident

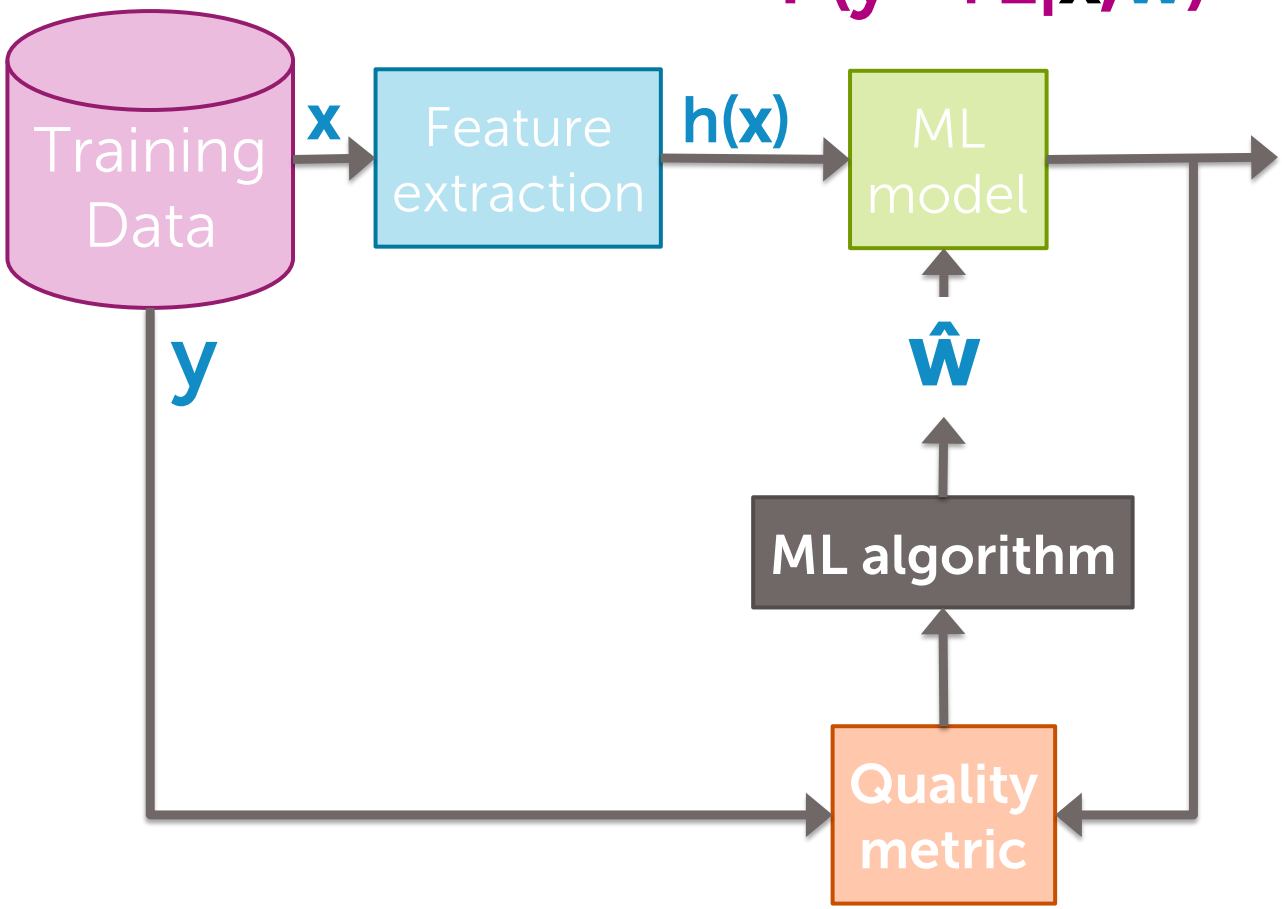
very natural uncertainty region

©2015-2016 Emily Fox & Carlos Guestrin

Finding best L_2 regularized linear classifier with gradient ascent

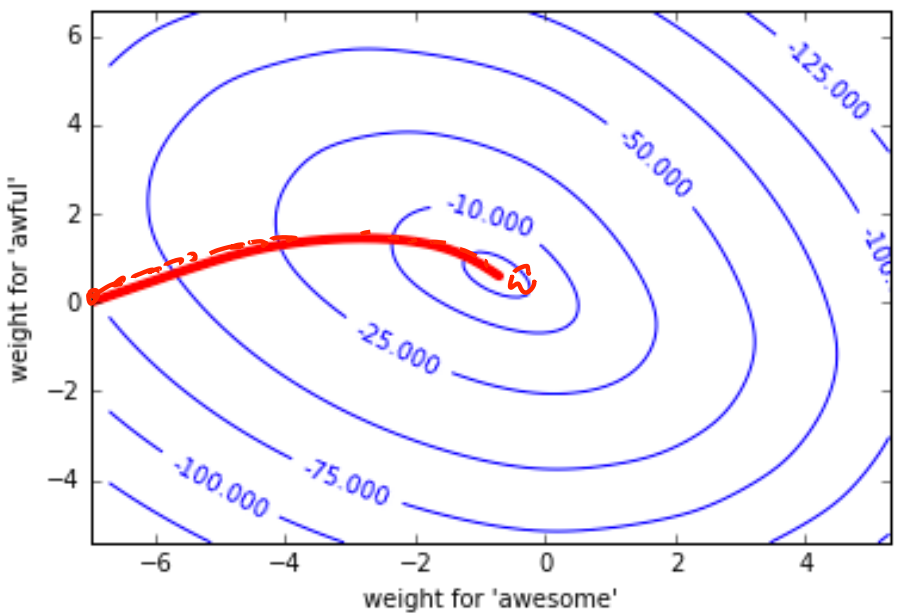
©2015-2016 Emily Fox & Carlos Guestrin

$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1+e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$



©2015-2016 Emily Fox & Carlos Guestrin

Gradient ascent



Algorithm:

while not converged

$$\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} + \eta \underbrace{\nabla \ell(\mathbf{w}^{(t)})}$$

need the gradient of regularized log likelihood

©2015 Emily Fox & Carlos Guestrin

Gradient of L_2 regularized log-likelihood

Total quality =

measure of fit - measure of magnitude of coefficients

$$\underbrace{\ell(\mathbf{w})}_{\text{measure of fit}} \qquad \underbrace{\lambda\,||\mathbf{w}||_2^2}_{\text{measure of magnitude of coefficients}}$$

$$\text{Total derivative} = \frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} - \lambda \frac{\partial ||\mathbf{w}||_2^2}{\partial \mathbf{w}_j}$$

©2015 Emily Fox & Carlos Guestrin

Derivative of (log-)likelihood

Sum over data points

Feature value

Difference between truth and prediction

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w})\Big)$$

Derivative of L_2 penalty

$$\frac{\partial ||\mathbf{w}||_2^2}{\partial \mathbf{w}_j} = \frac{\partial}{\partial w_j}\left[w_0^2 + w_1^2 + w_2^2 + \cdots + w_j^2 + \cdots + w_D^2\right] = 2w_j$$

©2015-2016 Emily Fox & Carlos Guestrin

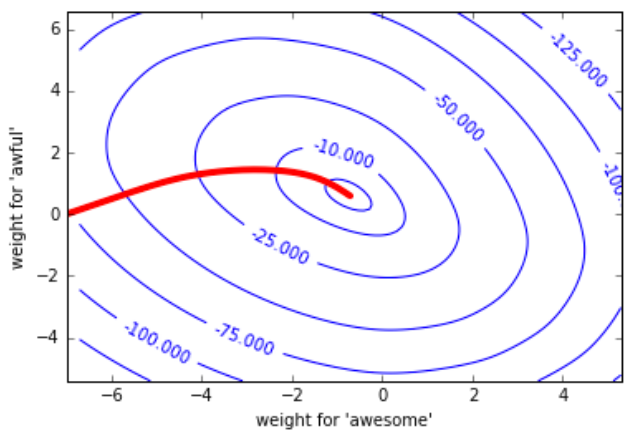
Understanding contribution of L_2 regularization

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} \overbrace{- 2\lambda \mathbf{w}_j}^{\text{Term from } L_2 \text{ penalty}}$$

	$-2\lambda w_j$	Impact on w_j
$w_j > 0$	< 0	decreases w_j => w_j becomes closer to 0
$w_j < 0$	> 0	increases w_j => w_j becomes closer to 0

©2015-2016 Emily Fox & Carlos Guestrin

Summary of gradient ascent for logistic regression with L_2 Regularization



init $\mathbf{w}^{(1)}=0$ (or randomly, or smartly), t=1

while not converged:

for j=0,…,D

$$\text{partial[j]} = \sum_{i=1}^{N} h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i=+1] - P(y=+1 \mid \mathbf{x}_i, \mathbf{w}^{(t)})\Big)$$

$$\mathbf{w}_j^{(t+1)} \leftarrow \mathbf{w}_j^{(t)} + \eta\,(\text{partial[j]} - 2\lambda\,\mathbf{w}_j^{(t)})$$

$$t \leftarrow t+1$$

step size

$\frac{\partial \ell(w)}{\partial w_j}$

only change !!

©2015 Emily Fox & Carlos Guestrin

Sparse logistic regression with L_1 regularization

©2015-2016 Emily Fox & Carlos Guestrin

Recall sparsity (many $\hat{w}_j$=0) gives efficiency and interpretability

Efficiency:

- If size($\mathbf{w}$) = 100B, each prediction is expensive
- If $\hat{\mathbf{w}}$ sparse, computation only depends on # of non-zeros

many zeros

$$\hat{y}_i = sign\left(\sum_{\hat{\mathbf{w}}_j \neq 0} \hat{\mathbf{w}}_j h_j(\mathbf{x}_i)\right)$$

Interpretability:

- Which features are relevant for prediction?

©2015 Emily Fox & Carlos Guestrin

Sparse logistic regression

Total quality =

measure of fit - measure of magnitude of coefficients

$\ell(\mathbf{w})$ $\quad$ $||\mathbf{w}||_1 = |w_0| + \ldots + |w_D|$

Leads to **sparse** solutions!

L_1 regularized logistic regression

©2015 Emily Fox & Carlos Guestrin

L_1 regularized logistic regression

Just like L2 regularization, solution is governed by a continuous parameter λ

$$\ell(\mathbf{w}) - \lambda||\mathbf{w}||_1$$

tuning parameter = balance of fit and **sparsity**

If λ=0:

No regularization → standard MLE solution

If λ=∞:

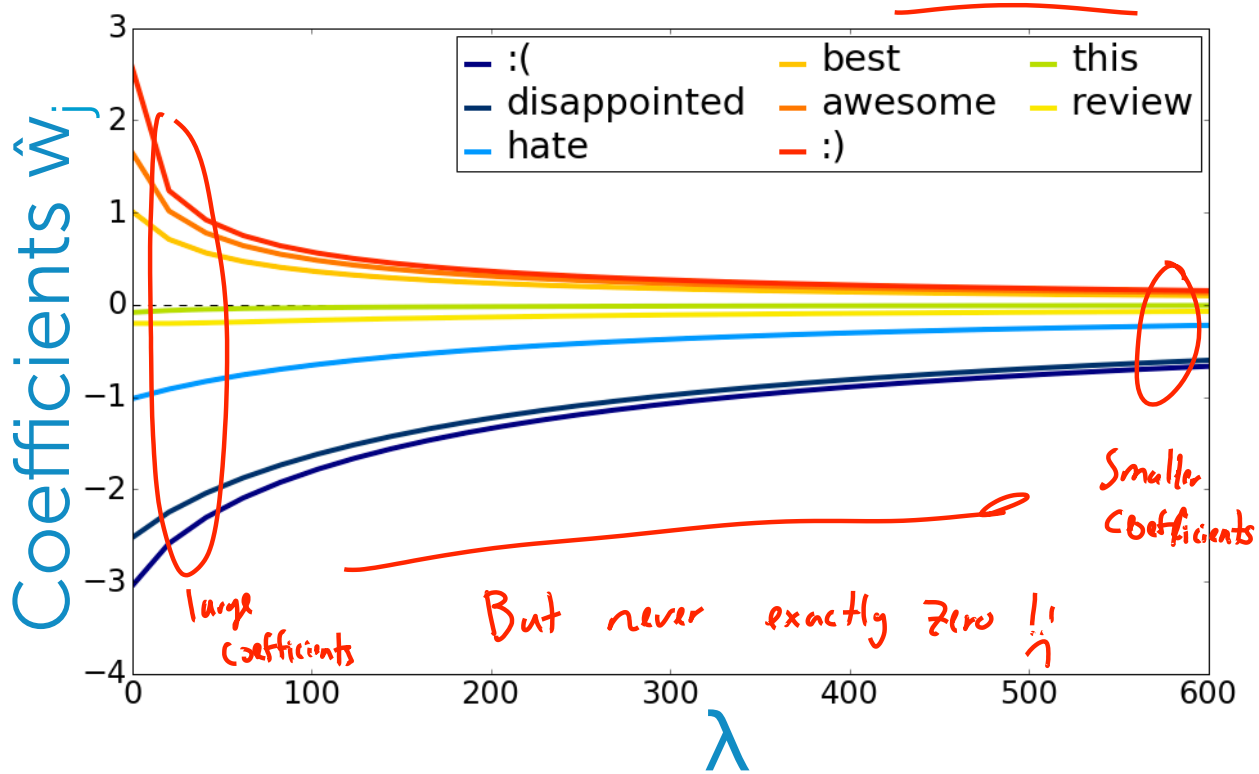
all weight is on regularization → $\hat{w}=0$

If λ in between:

Sparse solutions: Some $\hat{w}_j \neq 0$, many other $\hat{w}_j = 0$

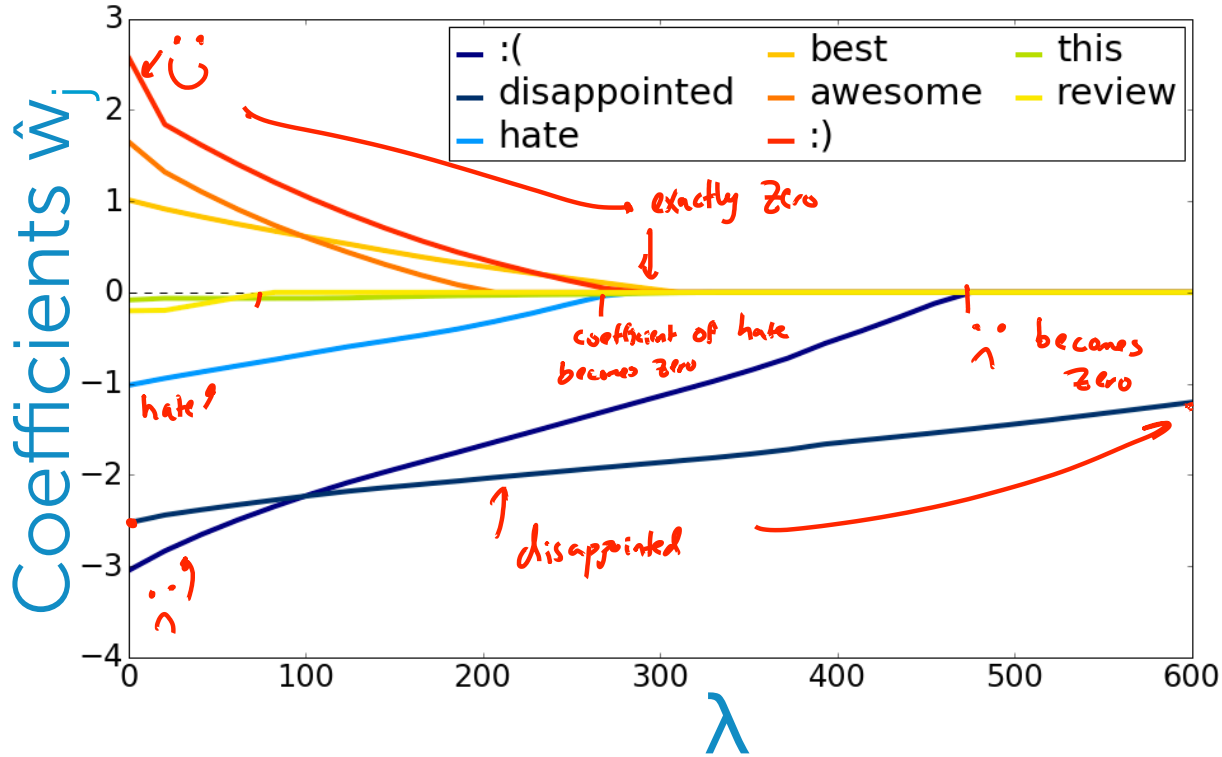
©2015 Emily Fox & Carlos Guestrin

Regularization path – L_2 penalty



©2015 Emily Fox & Carlos Guestrin

Regularization path – L_1 penalty



©2015 Emily Fox & Carlos Guestrin

Summary of overfitting in logistic regression

©2015-2016 Emily Fox & Carlos Guestrin

What you can do now…

- Identify when overfitting is happening
- Relate large learned coefficients to overfitting
- Describe the impact of overfitting on decision boundaries and predicted probabilities of linear classifiers
- Motivate the form of L_2 regularized logistic regression quality metric
- Describe what happens to estimated coefficients as tuning parameter λ is varied
- Interpret coefficient path plot
- Estimate L_2 regularized logistic regression coefficients using gradient ascent
- Describe the use of L_1 regularization to obtain sparse logistic regression solutions

©2015-2016 Emily Fox & Carlos Guestrin